

BASANT BHATTARAI

SENIOR DATA & AI ENGINEER / DATA PLATFORM & DATABASE SYSTEMS

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PROFILE

Senior data and AI engineer with nine years in software engineering, from Python backend services and PostgreSQL schemas to production data platforms, database reliability, and governed AI workflows. I focus on the seams that decide whether a system remains trustworthy: data contracts, observability, replay, recovery, and typed boundaries for automated decisions.

CORE STRENGTHS

Platform ownership Consumer-facing reliability, governance, recovery, and operational readiness.	Data engineering Batch and streaming pipelines, orchestration, modeling, and change-safe processing.
Database systems Transactional and analytical modeling, replication, recovery, and performance work.	Governed AI Structured output, provenance, evaluation, bounded tools, and human review paths.
Technical leadership Design review, mentoring, runbooks, and making system ownership transferable.	Distributed work Long-term collaboration across European and US time zones from Kathmandu.

EXPERIENCE

Senior Data Engineer | UXCam

Feb 2024 - present

- Own reliability and governance outcomes across production data and AI-assisted workflows, including clear service expectations and incident-ready runbooks.
- Design typed, validated, traceable boundaries before automated output can change another product workflow.
- Lead architecture and operational reviews across processing, storage, serving, observability, and recovery.
- Mentor engineers through pairing and review, with an emphasis on transferable ownership and clear decisions.

Data Engineer | UXCam

Feb 2020 - Feb 2024

- Built and operated batch and streaming pipelines, then expanded into data modeling, orchestration, storage, and analytical serving.
- Improved change and recovery safety by designing for schema evolution, replay, late data, and backfills.
- Tuned processing and query workloads by starting with access patterns, measurement, and data layout.

Project Leader | SVCET, India

2019 - 2020

- Led a small team delivering an NLP dialogue-system project across academic and industry stakeholders.

Software Developer | SV Technology, India

2017 - 2019

- Built Python backend services, PostgreSQL schemas and reports, and contributed to CI and early pipeline work.

INDEPENDENT WORK

ClickHomes - personal product

Founder and sole engineer. Public evidence includes 90+ versioned SQL migrations and a documented approach to contracts, idempotency, schema evolution, and reviewable AI-assisted decisions.

Multi-engine HA lab

Eight reproducible failover scenarios on hardware I control. Published results include PostgreSQL 18 replay-lag and Redis Sentinel promotion measurements, with the rig and limitations stated.

TECHNICAL PRACTICE

Programming

Python, SQL, Bash; application and automation work with modern JavaScript/TypeScript stacks.

Data processing

Apache Spark / PySpark, Kafka, Airflow, dbt, Iceberg, streaming and change-data-capture patterns.

Databases

PostgreSQL, ClickHouse, Trino; production or lab work with Redis, MongoDB, Citus, TimescaleDB, ScyllaDB, MariaDB/Galera, and SolrCloud.

AI systems

LangGraph, LangChain, CrewAI, Google-ADK, MCP, Pydantic, retrieval, evaluation, provenance, and structured-output design.

Infrastructure

Kubernetes, Docker, Terraform, Helm, GitHub Actions, Prometheus, Grafana, OpenTelemetry.

Cloud

Hands-on delivery across AWS, GCP, and Azure services for data, compute, storage, orchestration, and AI workloads.

EDUCATION AND LANGUAGES

Bachelor of Technology in Computer Science and Engineering

JNTUA College of Engineering, Anantapur, India | 2015 - 2019

English: C1 | Nepali: native

PUBLIC DISCLOSURE BOUNDARY

This public resume names employers and responsibilities but intentionally excludes employer architecture, customer information, operating scale, security posture, and internal performance figures. Independent metrics refer only to personal work with a stated measurement method.

Full public resume and evidence: technobasant.github.io/resume/